Coarse-to-Fine High-Order Network for Hyperspectral and LiDAR Classification

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***Abstract*— The fusion of hyperspectral and light detection and ranging (LiDAR) data could significantly improve land-cover classification performance. Most existing feature fusion methods focus on "late fusion" or "halfway feature interaction fusion" methods, which treat LiDAR and hyperspectral data as model inputs while overlooking the redundancy between hyperspectral imagery (HSI) and LiDAR features. In addition, the unique characteristics of HSI and LiDAR data, combined with the complexity of land-cover backgrounds, make it challenging to accurately describe their properties. High-order deep features, as a deep statistical representation, could effectively capture the statistical characteristics of these land covers. Based on this, this article focuses on the unique characteristics of HSI and LiDAR data, as well as the distinguishability of features, and designs a progressive hyperspectral and LiDAR collaborative classification method, named coarse-to-fine high-order network (CHNet). In the coarse stage, HSI data redundancy reduction and LiDAR feature reconstruction are performed in either the frequency domain or the spatial domain to ensure the effectiveness of subsequent feature fusion. The fine stage focuses primarily on selective feature fusion and discriminability enhancement, introducing gating mechanisms, deep expert systems, and high-order feature statistics. This approach enhances the discriminability of the fused features while simultaneously reducing their dimensionality. The proposed method, based on a "redundancy removal-feature learning" mechanism, captures more effective deep features by accounting for the different imaging mechanisms of multisource data and the complex background of land covers, ultimately improving the effectiveness of land-cover classification. Experimental results on three public datasets and one self-constructed dataset demonstrate that CHNet achieves superior performance. The code is available at https://github.com/RSIP-NJUPT/CHNet.**

***Index Terms*— Classification, feature fusion, hyperspectral imagery (HSI), light detection and ranging (LiDAR), redundancy.**

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I. Introduction

REMOTE sensing image data are widely used in fields such as urban planning and environmental monitoring for land-cover classification. Recently, with the development of remote sensing technology, it has become possible to acquire multisource remote sensing data of the same scene. Hyperspectral image data are characterized by high spectral resolution [1], [2], [3], which allows for accurate reflection of the spectral characteristics of different land covers [4], [5], [6]. However, hyperspectral imagery (HSI) data also suffer from issues such as spectral information redundancy and the phenomenon of different spectra for the same object, which affects land-cover classification [7], [8], [9], [10]. Light detection and ranging (LiDAR) data can acquire digital surface model (DSM) information [11], [12], providing the 3-D height of land covers. Therefore, LiDAR data can serve as a complement to hyperspectral data.

In terms of data characteristics, hyperspectral and LiDAR data collaborative classification faces the following issues.

1) Hyperspectral data have spectral channels ranging from dozens to even hundreds. Ineffective or redundant spectral information can impact the characterization of ground objects. Spectral band selection or spectral redundancy reduction methods are effective solutions.
2) The spatial resolutions of HSI and LiDAR are typically different, with HSI and LiDAR focusing on spectral and spatial features, respectively. Cross-modal feature

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fusion can affect the representation of land covers. Although deep neural features can effectively overcome the differences in data characteristics such as resolution between HSI and LiDAR data, the key challenge lies in how to selectively fuse distinguishable deep features.

Currently, the PCA method is employed for the redundancy reduction in HSI data [13], [14], which eliminates the problem of linear features. However, for HSI data with special imaging mechanisms, the PCA method has limitations while dealing with highly nonlinear feature redundancy or the influence of noise. Nowadays, deep neural networks have demonstrated excellent performance in the collaborative classification of HSI and LiDAR, particularly in the areas of spectral band selection and feature redundancy reduction. This is mainly due to the strong parameter learning and model generalization capabilities of deep neural networks. For example, Yang et al. [15] proposed a LiDAR-guided cross-attention fusion network for alleviating the high-dimensionality and redundancy challenges in HSI and LiDAR collaborative classification. Moreover, a multiscale head selection transformer network [16] was proposed to comprehensively explore detailed features across the spatial and spectral dimensions of HSI and LiDAR data. Some of the aforementioned methods, after utilizing PCA for feature dimensionality reduction, focus on nonredundant feature learning and spectral selection for subsequent global features. The methods used primarily include spectral attention mechanisms, improvements to the self-attention mechanism in transformers, and others. However, for HSI and LiDAR data, both data redundancy and the effectiveness of feature learning need to be considered. Therefore, performing feature learning from coarse to fine and effectively embedding data nonredundancy and feature nonredundancy methods into the network's coarse feature learning stage offer significant advantages for subsequent deep feature enhancement and fusion. Recent advancements in frequency-domain methods have demonstrated significant progress in image processing, particularly in feature modeling and fusion for both RGB and remote sensing images. For RGB images, Kong et al. [17] proposed a frequency-domain-driven transformer method, which significantly improved image deblurring performance through a frequency-domain self-attention solver (FSAS) and a discriminative feedforward network (DFFN), providing new insights into frequency-domain feature modeling. Ma et al. [18] designed a frequency attention denoising network (FADNet), leveraging frequency-domain attention mechanisms to fully characterize feature correlations across a broad frequency spectrum. This approach addressed the low-frequency bias of traditional CNN denoisers, significantly enhancing denoising performance and model generalization. For frequency-domain methods applied to remote sensing images, Yang et al. [19] introduced a multifrequency graph convolutional network with cross-modality mutual enhancement (MFGCN-CME). By extracting low- and high-frequency features of hyperspectral and LiDAR data in parallel through an adaptive multifrequency graph feature learning module and enhancing intermodality information via a bipartite graph enhancement learning module, the method significantly improved multisource remote sensing data classification performance. Zhao et al. [20] developed a fractional Fourier image transformer (FrIT), which effectively extracts global contextual and local semantic information by connecting multisource features in fractional domains, outperforming traditional CNNs and transformers on multiple datasets. Zeng et al. [21] proposed a cross-modal hierarchical frequency fusion network (HFNet), which adaptively fuses multilevel features of hyperspectral and LiDAR data in the frequency domain and incorporates a frequency consistency loss (FCL) to mitigate modality discrepancies, further enhancing classification performance. These methods highlight the unique advantages of frequency-domain feature modeling and cross-modality fusion, offering new research directions for multisource remote sensing data classification. Thus, we propose to design feature nonredundancy learning methods specifically for HSI and LiDAR data to enhance feature discriminability.

Multimodal feature fusion can utilize the rich spectral information provided by HSI and the precise spatial structural information from LiDAR, thereby improving land-cover classification. Currently, most feature fusion methods involve dual-branch feature fusion, local–global feature fusion, and others. These approaches could capture the complementary information between HSI and LiDAR data. Dong et al. [22] designed a multibranch feature fusion network, including a self- and cross-guided attention mechanism and an effective spectral–spatial-elevation feature learning style. Feng et al. [23] paid much attention to multisource information fusion and integrated spectral–spatial–elevation features via dual-channel graph convolutional network. The aforementioned methods involve feature selection and multimodal feature fusion at the feature level, but, in most cases, dynamic adaptive learning of module parameters offers a better characterization of ground object features for complex HSI and LiDAR data. In addition, multimodal feature selection and fusion can serve as a refined feature learning stage for HSI and LiDAR. Therefore, dynamic adaptive parameter learning and improvement of feature discriminability will be given priority in this stage.

Based on the above analysis, we design a coarse-to-fine two-stage HSI and LiDAR collaborative classification network, called a coarse-to-fine high-order network (CHNet). Our main contributions include the following.

1) During the coarse feature learning stage, we design a spectral redundancy removal (SRR) module and a spatial reconstruction (SR) module in the HSI and LiDAR branches, respectively. These two modules perform spectral feature redundancy removal in the frequency domain and LiDAR spatial feature redundancy removal in the spatial domain.
2) During the fine feature learning stage, we develop a dynamic feature selection and fusion (DFSF) module, which integrates a gating network and an expert system to adaptively perform selective feature fusion, thereby enhancing the feature learning capability in complex scenarios with intricate land-cover structures.
3) We introduce a factorized high-order feature learning (FHFL) module, which reduces the dimensionality of the fused features while enhancing their discriminability.

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In addition, experimental results on three public datasets and one self-construct dataset have validated the effectiveness of the proposed network.

The rest of this article is structured as follows. Section II offers a brief overview of CNN-transformer networks and cross-modal feature fusion methods. Section III details the proposed CHNet. Section IV discusses and analyses experimental results on three publicly available datasets and one self-collected dataset. Finally, Section V presents future work and concludes this article.

II. Related Work

A. CNN-Transformer Network

Driven by the success of convolutional neural networks (CNNs) [24], [25] and transformer, several researchers have also explored CNNs and transformer network [20] for hyperspectral and LiDAR classification [26], [27]. CNNs have strong local feature extraction capabilities in HSI and LiDAR feature learning but are insufficient in modeling global dependencies. Transformers excel at capturing long-range dependencies and global information, making them particularly suitable for data with high spectral dimensions or requiring global modeling. The combination of CNNs and transformer can effectively capture the local–global features of HSI and LiDAR. Recently, Sun et al. [28] employed a multiscale 3-D and 2-D mixed-grouped CNNs to capture local spectral–spatial–elevation features; moreover, this method inserted a DConvformer module into a conventional transformer block to extract morphological features and feature fusion. In particular, Song et al. [29] followed an encoder–decoder architecture and proposed a transformer-based spectral encoder and CNN-based spatial encoder for efficient spectral and spatial feature learning. Other related works include the hierarchical CNN and transformer (HCT) [30], the local information interaction transformer [31], the spectral–spatial morphological attention transformer for hyperspectral feature learning [32], and so on. Currently, some methods use CNN-transformer networks for HSI and LiDAR land-cover classification, such as the fractional Fourier transformer [33], the multiscale head selection transformer [16], and the multimodal informative ViT [34].

The aforementioned method can effectively capture both local and global features of HSI and LiDAR. Most methods improve either the design of convolutional layers or the design of transformer self-attention mechanisms to enhance the feature discriminability. However, this approach fails to address the feature redundancy present in HSI data and the inherent limitations of PCA for dimensionality reduction. Therefore, the proposed network focuses on reducing redundancy and reconstructing features in HSI and LiDAR data to improve classification accuracy.

B. Cross-Modal Feature Fusion

A cross-modal enhanced CNNs and transformer module [35], [36] is developed to effectively augment the fused interactive local and global features derived from HSI and LiDAR data cubes. Li et al. [37] implemented a bilateral modulation strategy, with feature fusion applied at every stage of the proposed network, effectively reducing semantic drift between features. For the feature interaction of coastal wetland classification, the HSI and LiDAR features interacted in channels and pixels [38]; furthermore, the sequence-to-sequence fusion stage and the low-frequency residual enrichment stage are utilized to complete the sequence level fusion of features and features merge, respectively. Shen et al. [39] designed a cross-layer and cross-interaction module and a feature concatenation module to enhance and merge the cross-modal features. Other related works include the uncertain category-aware fusion network (UCAFNet) [40], the multiple information collaborative fusion network (MICF-Net) [41], the dynamic scale hierarchical fusion network (DSHFNet) [42], the graph-attention-based multimodal fusion network (GAMF) [43], the attention-based feature fusion module [44], and the bilateral interactive hierarchical adaptive fusion network (BIHAF-Net) [45].

In cross-modal feature fusion, most methods employ attention mechanisms for feature interaction and fusion. This approach assigns different weights to features from different channels, but some methods fail to consider the dynamic parameter learning of the modules and the impact of subsequent enhanced feature fusion on the effectiveness of land-cover classification. Therefore, our method will utilize dynamic feature fusion learning and feature enhancement to precisely capture the characteristics of land covers.

III. Methodology

Fig. 1 illustrates the overall framework of our coarse-to-fine two-stage HSI and LiDAR collaborative classification network. In the coarse stage, redundancy in hyperspectral and LiDAR data is addressed separately. Specifically, the SRR module is designed to filter out redundant data in the frequency domain for hyperspectral data, while the SR module performs redundancy filtering in the spatial domain for LiDAR data. In the fine stage, a feature fusion and enhancement module is introduced to select and enhance the learned deep fusion features. As shown in Fig. 1, the complete framework is composed of five main modules: the SRR module, the SR module, the multiscale feature extraction (MFE) module, the DFSF module, and the FHFL module.

In the source data, $\mathbf{X}_H \in \mathbb{R}^{h\times w\times l}$ represents the hyperspectral cube, and $\mathbf{X}_L \in \mathbb{R}^{h\times w}$ corresponds to the LiDAR elevation information for the same geographical area. Here, h and w denote the original height and width of the data, while l indicates the number of spectral channels. The data ARE then split into smaller patch cubes, $\mathbf{X}_H^S \in \mathbb{R}^{s\times s\times l}$, and $\mathbf{X}_L^S \in \mathbb{R}^{s\times s}$, where s represents the size of each patch cube. Each patch corresponds to the class of its central pixel, with edge pixels being filled via padding. This results in a total of $m \times n$ patch cubes. After discarding the patch cubes representing class 0, the remaining patch cubes are split into training and testing sets, which are then used as input data.

A. SRR Module

In the coarse stage, we designed an SRR module specifically for the preliminary screening and removal of redundant

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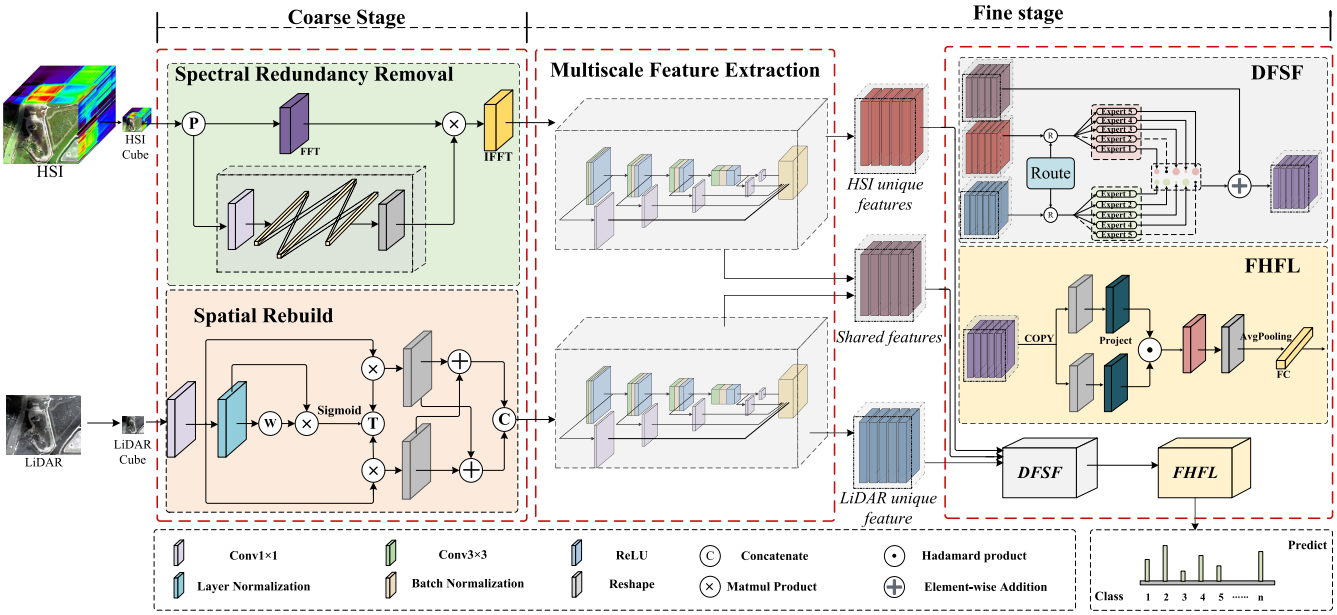


Fig. 1. Overall framework of CHNet for HSI and LiDAR classification, including the SRR module, the SR module, the MFE module, the DFSF module, and the FHFL module.

information in hyperspectral data. This module operates by adaptively filtering out irrelevant spectral information in hyperspectral images within the frequency domain, effectively reducing redundancy. Specifically, the input patches are transformed into the frequency domain, where a dynamic filter is predicted to adaptively reduce redundant spectral information in the HSI modality [46].

First, the hyperspectral data $\mathbf{X}_H^S$ are fed into the SRR module. Following dimensionality reduction with PCA, the data are processed with a discrete Fourier transform (DFT) to produce the corresponding frequency-domain images $\mathbf{F}_{\text{hsi}}$. Subsequently, the filter prediction network dynamically generates filters to reduce the redundant spectrum

$$\mathbf{F}_{\text{hsi}} = \text{DFT}\big(\text{PCA}\big(\mathbf{X}_H^S\big)\big). \tag{1}$$

The entire image is initially processed through a convolution operation, and then, the resulting output is flattened and passed into a multilayer perceptron (MLP) layer to generate feature vectors $\mathbf{V}_{\text{hsi}}$. Based on these feature vectors, a top-k operation is performed to generate binary filters. The top-k operation sets the positions of the top-k values to 1, while the remaining positions are set to 0, and the result is reshaped to match the original image size, producing a mask $\mathbf{M}_{\text{hsi}}$

$$\mathbf{V}_{\text{hsi}} = \text{Flatten}(\text{Conv}_{3\times3}(\mathbf{F}_{\text{hsi}})) \tag{2}$$

$$\mathbf{M}_{\text{hsi}} = \text{Reshape}(\text{topK}(\text{MLP}(\mathbf{V}_{\text{hsi}}))). \tag{3}$$

Finally, $\mathbf{M}_{\text{hsi}}$ is applied to the Fourier-transformed image for filtering. By performing an inverse DFT (IDFT), the image is transformed back, resulting in the final image with the redundant spectra removed. This process effectively eliminates irrelevant spectral information in the frequency domain, enhancing image quality and laying a solid foundation for subsequent processing steps. The dynamic prediction of the filter ensures that the filtering process is adaptive and targeted, allowing adjustments based on the content of the image

$$\mathbf{X}'_H = \text{IDFT}(\mathbf{M}_{\text{hsi}} \otimes \mathbf{F}_{\text{hsi}}). \tag{4}$$

B. SR Module

The SR module represents another component of the coarse stage, designed to enhance feature representation while reducing redundant information interference, thus providing better inputs for subsequent processing [47]. Specifically, after enhancing the spatial representation through convolution, the module employs normalization and threshold mechanisms to perform feature selection and enhancement using separation and reconstruction operations.

First, the LiDAR image data $\mathbf{X}_L$ are passed through the SR module via a convolution operation to produce $\mathbf{X}_L^{\text{Conv}}$, which is then fed into a layer normalization (LN) layer. Simultaneously, the scaling factor $\mathbf{R}_w$ and the threshold in the LN layer are used as benchmarks to evaluate the feature information. Then, the process can be described as follows:

$$\mathbf{X}_L^{\text{out}} = \text{LN}\big(\mathbf{X}_L^{\text{conv}}\big) = \alpha \times \frac{\mathbf{X}_L - \mu_L}{\sqrt{\sigma^2 - \varepsilon}} + \beta \tag{5}$$

where α and β are learnable parameters. μ and σ^2 represent the mean and standard deviation of X_L, respectively, and ε is a small positive constant added to ensure numerical stability during division. The feature's importance is reflected by calculating the ratio of α in the total sum. The weights of the feature maps, which are assigned different values, are then mapped to the range of (0, 1) by a sigmoid function

$$\mathbf{R}_w = \frac{\alpha_i}{\sum_{j=1}^{c}\alpha_j}, \quad i, j = 1, 2, \ldots, c. \tag{6}$$

Through the threshold mechanism, where the threshold is set to 0.5, the mapped weights are compared. Weights greater than the threshold are set to 1, resulting in the information-bearing

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weight W_1, while weights smaller than the threshold are also set to 0, yielding the noninformation-bearing weight W_2. The calculation is given as follows:

$$\mathbf{W} = \text{Route}\left(\text{Sigmoid}\left(\mathbf{R}_w \otimes \mathbf{X}_L^{\text{out}}\right)\right). \tag{7}$$

Finally, $\mathbf{X}_L^{\text{conv}}$ is multiplied by $\mathbf{W}_1$ and $\mathbf{W}_2$ separately, producing two characteristics: one more informative feature X_1 and another with minimal information $\mathbf{X}_2$. $\mathbf{X}_2$ can be considered redundant information, which is then aggregated by blocks to enhance feature representation and reduce redundancy

$$\begin{cases} \mathbf{X}_1 = \mathbf{X} \otimes \mathbf{W}_1, \mathbf{X}_2 = \mathbf{X} \otimes \mathbf{W}_2 \\ \mathbf{X}_1^1, \mathbf{X}_1^2 = \text{Split}(\mathbf{X}_1) \\ \mathbf{X}_2^1, \mathbf{X}_2^2 = \text{Split}(\mathbf{X}_2) \\ \mathbf{X}_L{}' = \left(\mathbf{X}_1^1 \oplus \mathbf{X}_2^2\right) \cup \left(\mathbf{X}_2^1 \oplus \mathbf{X}_1^2\right) \end{cases} \tag{8}$$

where $\otimes$ and $\oplus$ denote the elementwise multiplication and elementwise summation, respectively, and $\cup$ signifies the concatenation. Split $(\cdot)$ refers to dividing the feature into two parts along the channel dimension.

C. Mutiscale Feature Extraction

The feature extraction module is a critical component of the fine processing stage, designed to play a pivotal role in hierarchical feature extraction from input images. Its structure is identical for both hyperspectral and LiDAR data, allowing for multilevel feature abstraction. At lower levels, it captures fine-grained details, while, at deeper levels, it identifies broader, high-level patterns. This progressive feature extraction is essential for tasks such as semantic segmentation, where understanding both local and global image contexts is crucial.

The module employs multiple convolution layers and batch normalization to ensure stable learning and mitigate the vanishing gradient problem and incorporates a feature pyramid network (FPN) [48] to merge multiscale features through top-down pathways and lateral connections. This approach is especially beneficial for image classification, where land covers at multiple scales are the key to accurate classification. By constructing a feature pyramid, the module effectively balances fine details with contextual information, enhancing precision and recall. In addition, the use of a Laplacian operator strengthens the model's ability to capture edge information, which is critical for distinguishing land covers in HSI and LiDAR data. Edge detection, based on the Laplacian filter, enables the model to focus on boundaries, improving the accuracy of object boundary segmentation—an area where traditional convolution layers may struggle.

Specifically, the module begins with four 2-D convolution layers, each with a kernel size of 3×3 and channels set to [64, 128, 256, 512]. Each convolution is followed by batch normalization and an ReLU activation function. The module $M\text{Conv}$ processes deredundant images $\mathbf{X}'_H$ and $\mathbf{X}'_L$, applying them to the first convolutional layer, followed by batch normalization and ReLU activation. This process is repeated across subsequent layers, with each layer increasing the number of channels and applying progressively more complex filters. The outputs of all four layers $\mathbf{F}_H$ and $\mathbf{F}_L$ are stored and returned as a list of feature maps, representing hierarchical feature abstractions at different depths of the network for both hyperspectral and LiDAR data

$$\mathbf{F}_H, \mathbf{F}_L = M\text{Conv}\left(\mathbf{X}'_H, \mathbf{X}'_L\right). \tag{9}$$

The features are subsequently processed by the FPN, which applies a 1×1 convolution to adjust the channels of all scales to 64

$$\mathbf{F}_H^{\text{fpn}}, \mathbf{F}_L^{\text{fpn}} = \text{FPN}\left(M\text{Conv}\left(\mathbf{X}'_H, \mathbf{X}'_L\right)\right). \tag{10}$$

Finally, the features $\mathbf{X}'_H$ and $\mathbf{X}'_L$ processed by the Laplacian convolution are concatenated to the corresponding scale-specific features, ensuring that the enhanced model captures both detailed and contextual information

$$H_{\text{lap}} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & +8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \tag{11}$$

$$\begin{cases} \mathbf{E}_H = H_{\text{lap}} * \mathbf{X}'_H \\ \mathbf{E}_L = H_{\text{lap}} * \mathbf{X}'_L \end{cases} \tag{12}$$

where H_{lap} is the convolution kernel and * is the convolution operation. $\mathbf{E}_H$ and $\mathbf{E}_L$ are edge features of hyperspectral and LiDAR images, respectively,

$$\mathbf{X}_{\text{shared}} = \left(\mathbf{F}_H^{\text{fpn}} + \mathbf{F}_L^{\text{fpn}}\right) \oplus \left(\mathbf{E}_H + \mathbf{E}_L\right) \tag{13}$$

where $\oplus$ represents the concatenation operation. First, the features $\mathbf{F}_H^{\text{fpn}}$ and $\mathbf{F}_L^{\text{fpn}}$ from both modalities are added elementwise. Then, their edge features $\mathbf{E}_H$ and $\mathbf{E}_L$ are also summed. Finally, these two results are concatenated, yielding the final shared feature $\mathbf{X}_{\text{shared}}$.

D. DFSF Module

In the fine stage, we propose a DFSF module that adaptively fuses features based on their scale-specific characteristics. To achieve this, we design a scale-aware mixture of experts (MoEs) model, where each expert is specialized in processing modal-specific features at a particular scale [46]. The gating network, informed by the bimodal data, dynamically assigns weights to these experts, enabling effective feature fusion.

Specifically, after removing redundant information from the two modalities, we first apply average pooling to reduce spatial dimensions and flatten the data into 1-D vectors. These vectors are then fed into a multilayer perceptron (MLP) to predict the gating weights $\mathbf{w}_H$ and $\mathbf{w}_L$ for hyperspectral and LiDAR modalities, respectively. A softmax normalization is applied to ensure that the weights are within the range [0, 1], representing the contribution of each modality to the final fusion. The formalization is given as follows:

$$\mathbf{w}_H, \mathbf{w}_L = \text{Softmax}\left(\text{MLP}\left(\text{flatten}\left(\mathbf{X}'_H, \mathbf{X}'_L\right)\right)\right) \tag{14}$$

$$\mathbf{s}_1, \mathbf{s}_2 = \text{Route}(\mathbf{w}_H, \mathbf{w}_L) \tag{15}$$

where Route $(\cdot)$ denotes a threshold-based routing mechanism. Here, a predefined threshold $\mathbf{t}$ is used to compare the weights $\mathbf{w}_H, \mathbf{w}_L$. If a weight exceeds $\mathbf{t}$, the corresponding feature is considered significant and is assigned a value of 1; otherwise, it is set to 0. The resulting binary masks $\mathbf{s}_1$ and $\mathbf{s}_2$ are used to

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selectively activate the appropriate experts for further feature extraction. This ensures that only the most relevant features proceed to the next stage

$$\mathbf{F}'_H = \mathbf{E}_H(\mathbf{F}_H \cdot \mathbf{s}_1), \quad \mathbf{F}'_L = \mathbf{E}_L(\mathbf{F}_L \cdot \mathbf{s}_2) \tag{16}$$

where E_H and E_L represent multiscale expert networks tailored for hyperspectral and LiDAR data, respectively. These experts further refine the unique features of each modality, producing enhanced features F'_H and F'_L. The experts are designed with scale-specific convolutional layers to capture both local and global contextual information, ensuring robust feature extraction across different scales. The detailed structure of each scale-aware expert E is identical, primarily consisting of two 3×3 convolutional blocks.

Finally, the refined features are dynamically weighted and fused using the predicted gating weights $\mathbf{w}_H$ and $\mathbf{w}_L$. The fused features are combined with the shared features to produce the final multimodal representation

$$\mathbf{X}_{\text{fusion}} = \mathbf{X}_{\text{shared}} + \bigcup_{i=1}^{n} \left(\mathbf{w}_H^i \mathbf{F}_H^{i\prime} + \mathbf{w}_L^i \mathbf{F}_L^{i\prime} \right) \tag{17}$$

where X_{shared} represents the shared features extracted from both modalities, while $\mathbf{w}_H^i$ and $\mathbf{w}_L^i$ denote the gating weights for the ith scale-specific expert. The final fused features are obtained by concatenating the weighted outputs of all experts, ensuring a comprehensive representation of the multimodal data.

E. FHFL Module

This module aims to transform complex input data into efficient and compact feature representations by extracting and sparsifying key features, thereby enhancing the model's classification performance.

The module first applies bilinear projection and interaction to capture relationships and deep patterns within the input features. Sparse coding is then employed to reduce redundant information, retaining only the most discriminative features, which helps to minimize noise and irrelevant data interference. Pooling and normalization are subsequently applied to further compress and smooth the feature distribution, enhancing the model's robustness and generalization ability [49].

The main objective of this module is to efficiently extract and sparsify the input features, transforming the complex data structure into more representative features while preserving critical information. This process ultimately improves the accuracy of land-cover classification. The steps are given as follows.

Each of these two linear layers performs a different projection operation on the input features. The goal of this linear transformation is to map the input features into a new feature space ready for subsequent bilinear operations

$$\mathbf{U} = f^1\left(\mathbf{X}^i_{\text{fusion_}c}\right), \quad \mathbf{V} = f^2\left(\mathbf{X}^i_{\text{fusion}}\right) \tag{18}$$

$$\mathbf{P} = \mathbf{U} \odot \mathbf{V} \tag{19}$$

where $f^1(\cdot)$ and $f^2(\cdot)$ both represent projection operations and obtain the projected features $\mathbf{U}$ and $\mathbf{V}$. $\odot$ represents elementwise multiplication, applied to the results of two linear projections. This step can be understood as performing a bilinear interaction of features, which captures the interaction information between the two projected features, thereby providing a more comprehensive representation of the complex patterns within the input data.

Then, $\mathbf{P}$ is sparsified to reduce irrelevant or redundant information, and the sparsification is formulated as follows:

$$\mathbf{P} = \text{Sign}(\mathbf{P}) \odot \text{MAX}(|\mathbf{P}| - \beta/2, 0) \tag{20}$$

where $\text{Sign}(\cdot)$ denotes the extraction of the sign (positive or negative) of the input features, while $||$ represents the calculation of the absolute value of input features. The constant $\beta/2$ is then subtracted to control the degree of sparsification. This sparse representation enables the network to compress the feature space while preserving essential and meaningful information, thereby diminishing the influence of noise or irrelevant features

$$\mathbf{P}_{\text{avg}} = \text{AvgPool}(\mathbf{P}) \tag{21}$$

$$\mathbf{P}' = \sqrt{\text{ReLU}(\mathbf{P}_{\text{avg}})} - \sqrt{\text{ReLU}(-\mathbf{P}_{\text{avg}})} \tag{22}$$

$$\textbf{Output} = \text{MLP}\left(\text{Norm}(\mathbf{P}')\right). \tag{23}$$

The described process applies a double square root transformation to the features derived from bilinear pooling, effectively normalizing the output into a smoother and more nonlinear distribution. After this normalization, the features are then mapped to the classification output through an MLP, producing the final classification results.

IV. Experiments

A. Dataset

To evaluate the effectiveness of our proposed framework, CHNet, we utilized four datasets: three publicly available datasets, Houston2013, MUUFL, and Augsburg, and the Yancheng dataset provided by our research group.

1) Yancheng: The Yancheng dataset comprises HSI and LiDAR data, collected on October 15, 2022, at an altitude of 600 m over Yancheng City, China. The HSI data, obtained through the multiparameter airborne remote sensing system (MARSS) that is outfitted with a visible spectrum imaging spectrometer, initially comprised 249 spectral bands spanning wavelengths from 400 to 1000 nm. LiDAR data were recorded with the RIEGL VUX-240 laser scanner, achieving an average point density of 51.44 points per square meter. The two datasets exhibit identical dimensions of 400×500 pixels and possess a spatial resolution of 1 m. Collectively, the dataset encompasses nine distinct land-cover categories, comprising a total of 200 000 samples. The pseudocolor images synthesized from three bands of HSI, the LiDAR elevation map, and the ground-truth (GT) image are all presented in Fig. 2(a)–(c). Table I presents the number of training samples and testing samples allocated to each category.

2) Houston2013: The Houston2013 dataset, initially released as part of the IEEE GRSS Data Fusion Contest, remains publicly accessible. The dataset includes HSI and DSM data, which were collected in June 2012 by the National Center for Airborne Laser Mapping (NCALM) utilizing the ITRES CASI-1500 sensor. The HSI data encompass the University of Houston campus along with its adjacent urban

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TABLE I
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR YANCHENG DATA (BEST RESULTS ARE IN BOLD)

No.	Class (Train/Test)	CoupledCNN	S2ENET	MACN	EXViT	HCTNet	FDNet	HFSNet	M2FNet	MICF-Net	MS2CANet	**CHNet (ours)**
1	Suaeda glauca (20/2520)	97.18	97.46	94.08	96.27	94.56	98.25	88.09	91.34	42.30	**99.13**	95.87
2	Bare soil (20/30275)	72.09	73.05	75.2	65.66	24.66	71.65	75.14	67.62	72.33	74.47	**79.36**
3	Road (20/2956)	82.81	78.76	84.77	78.59	77.33	81.22	**86.46**	85.92	48.71	80.72	84.17
4	Buildings (20/87)	**100**	98.85	**100**	96.55	**100**	**100**	98.85	**100**	14.14	**100**	98.85
5	Water (20/49297)	85.72	85.16	77.28	78.78	77.85	**88.18**	85.80	72.09	85.57	84.41	82.24
6	Reed (20/42251)	53.04	68.78	67.44	62.35	22.43	68.97	69.72	**84.13**	77.64	64.92	73.01
7	S. alterniflora (20/58880)	56.94	66.62	80.73	78.26	82.81	73.80	64.61	66.08	**83.74**	70.48	78.60
8	Algae (20/3282)	88.18	85.59	83.18	89.58	89.52	**94.94**	82.05	88.54	34.89	93.57	93.08
9	Grass (20/10272)	68.33	54.49	71.29	67.28	69.83	71.38	62.15	64.90	52.50	67.70	**71.96**
	OA (%)	67.52	72.9	76.02	72.98	59.53	76.65	73.31	72.97	75.20	74.11	**78.63**
	AA (%)	78.26	78.75	81.55	79.26	71.00	83.15	79.21	80.07	56.87	81.71	**84.13**
	Kappa (%)	60.74	66.38	69.92	66.12	49.20	71.03	66.58	66.21	68.84	68.01	**73.32**

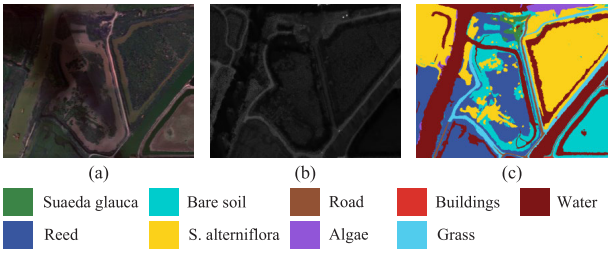


Fig. 2. Illustration of the Yancheng dataset. (a) Pseudocolor composite image based on bands 40, 60, and 120 for HSI. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

regions, comprising 144 spectral bands that range in wavelengths from 0.38 to 1.05 μm. The DSM data, represented by a single band, serve as a complementary modality to the HSI data. Both modalities exhibit a spatial resolution of 2.5 m and possess dimensions of 349 $\times$ 1905 pixels. The dataset comprises 15 unique land-cover classifications and contains a total of 15 029 labeled samples designated for both training and testing purposes. The pseudocolor images synthesized from three bands of HSI, the LiDAR elevation map, and the GT image are all presented in Fig. 3(a)–(c). Table II presents the training samples and testing samples allocated to each category.

3) MUUFL: The MUUFL dataset was gathered in November 2010 above the Gulf Park campus of the University of Southern Mississippi in Long Beach, MS, USA. The LiDAR data, acquired through the utilization of an ALTM sensor, comprise two raster datasets at a wavelength of 1.06 μm. The hyperspectral data, which were originally collected utilizing the ITRES CASI-1500 sensor, comprise 72 spectral bands. Following the exclusion of eight bands that were compromised by noise, a total of 64 spectral bands remained, with wavelengths spanning from 0.38 to 1.05 μm. Both the HSI and LiDAR datasets have a spatial size of 325 $\times$ 220 pixels. The dataset contains 53 687 labeled ground-truth samples that represent 11 distinct land-cover classes. The pseudocolor images of HSI, the LiDAR elevation map, and GT are all depicted in Fig. 4(a)–(c). Table III presents the number of training samples and testing samples allocated to each category.

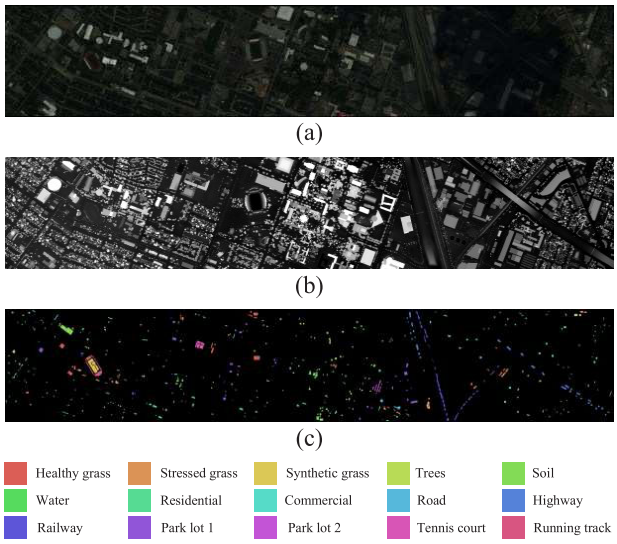


Fig. 3. Illustration of the Houston2013 dataset. (a) Pseudocolor composite image based on bands 64, 43, and 20 for HSI. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

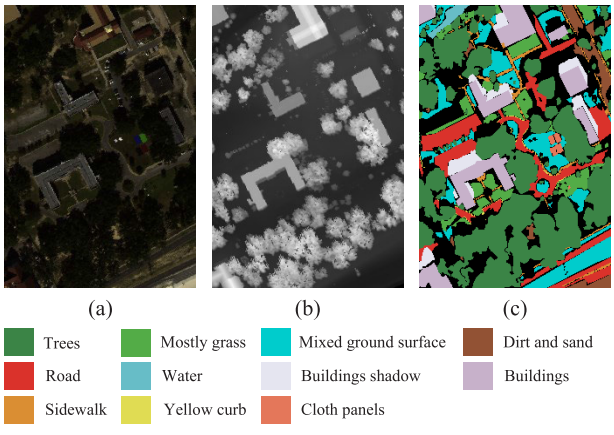


Fig. 4. Illustration of the MUUFL dataset. (a) Pseudocolor composite image based on bands 20, 15, and 5 for HSI. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

4) Augsburg: The Augsburg dataset was gathered above Augsburg, Germany, utilizing hyperspectral data obtained

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TABLE II
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR HOUSTON2013 DATA (BEST RESULTS ARE IN BOLD)

No.	Class (Train/Test)	CoupledCNN	S2ENET	MACN	EXViT	HCTNet	FDNet	HFSNet	M2FNet	MICF-Net	MS2CANet	ours
1	Healthy grass (20/1231)	85.70	83.51	94.31	96.51	95.69	97.32	97.80	94.80	**98.46**	88.38	97.89
2	Stressed grass (20/1234)	94.17	95.54	99.10	95.14	98.46	98.78	96.92	91.57	96.11	97.00	**99.76**
3	Synthetic grass (20/677)	91.14	98.82	99.85	93.94	99.85	97.49	**100**	99.55	96.58	98.08	99.56
4	Trees (20/1224)	69.53	99.26	99.34	99.84	97.96	99.84	99.59	97.05	96.71	96.65	**100**
5	Soil (20/1222)	99.92	**100**	98.36	98.94	**100**	**100**	99.59	98.69	98.87	**100**	**100**
6	Water (20/305)	97.05	**100**	99.01	99.67	98.03	96.39	94.75	95.41	99.34	99.34	**100**
7	Residential (20/1248)	92.07	94.84	93.42	91.67	**97.92**	96.23	96.71	95.19	95.84	84.38	96.95
8	Commercial (20/1224)	64.62	91.91	74.83	94.61	77.86	89.54	84.26	87.34	**98.18**	66.01	95.66
9	Road (20/1232)	93.34	91.56	93.01	88.88	83.52	84.50	81.66	92.04	92.59	68.34	**93.99**
10	Highway (20/1207)	72.99	92.29	97.84	87.90	97.18	**98.18**	94.61	95.52	89.15	97.76	95.77
11	Railway (20/1215)	33.50	97.45	**99.83**	95.97	93.74	96.30	97.53	93.41	94.44	75.80	94.65
12	Park lot 1 (20/1213)	77.33	92.00	88.95	90.68	92.66	**95.47**	90.76	90.93	94.58	92.66	94.55
13	Park lot 2 (20/449)	91.09	97.10	95.99	97.10	**99.55**	91.98	84.63	94.20	94.08	94.21	89.76
14	Tennis court (20/408)	99.75	**100**	**100**	**100**	**100**	**100**	**100**	98.53	**100**	**100**	**100**
15	Running track (20/640)	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	99.38	**100**	**100**
	OA (%)	81.30	94.98	94.77	94.65	94.53	95.91	94.89	94.39	95.81	88.64	**97.11**
	AA (%)	84.15	95.82	95.59	95.39	95.50	96.13	94.99	94.95	96.28	90.58	**97.23**
	Kappa (%)	79.77	94.57	94.35	94.22	94.09	95.58	94.47	93.93	95.47	87.73	**96.88**

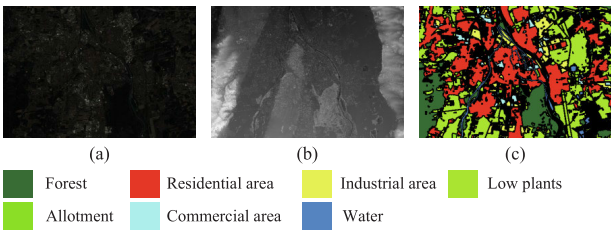


Fig. 5. Illustration of the Augsburg dataset. (a) Pseudocolor composite image based on bands 20, 15, and 10 for HSI. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

using the DAS-EOC HySpex sensor and LiDAR data acquired through the DLR-3K system [50]. To ensure uniformity in multimodal fusion, both datasets were standardized to a spatial resolution of 30 m. The HSI data comprise 180 spectral bands, encompassing a wavelength range from 0.4 to 2.5 μm, whereas the DSM data are represented by a single band. The dataset comprises 332 $\times$ 485 pixels and includes seven distinct land-cover categories. The pseudocolor images synthesized from three bands of HSI, the LiDAR elevation map, and the GT image are all presented in Fig. 5(a)–(c). Table IV presents the number of training samples and testing samples allocated to each category.

B. Experimental Setup

1) Configurations: The experiments pertaining to the proposed methodology, as well as various deep learning techniques, were executed on the PyTorch framework, utilizing an Ubuntu operating system equipped with an Intel i9-13900K central processing unit, an NVIDIA RTX 4090 graphics processing unit, and 32 GB of random access memory. The Adam optimizer was employed for network optimization. In the course of training, the batch size was established at 64, while the quantity of epochs was set to 100.

2) Evaluation Indicators: To objectively evaluate the classification performance of the proposed CHNet and other comparison networks, overall accuracy (OA), the Kappa coefficient (Kappa), and average accuracy (AA) are adopted as evaluation indicators.

C. Performance Comparison

To verify the validity of CHNet, several representative methods used for joint classification of HSI and LiDAR are used for comparison, which include CoupledCNN [51], the spatial–spectral cross-modal enhancement network (S2ENET) [52], the mixing self-attention and convolution network (MACN) [53], the extended ViT (ExViT) [26], a hierarchical CNN and transformer network (HCTNet) [30], the frequency-domain-based network (FDNet) [54], the height information guided hierarchical fusion-and-separation network (HFSNet) [29], the multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer (M2FNet) [55], MICF-Net [41], and the multiscale 3-D–2-D mixed CNN feature extraction and multisource data lightweight attention-free fusion network (MS2CANet) [56]. The efficacy of various comparative methodologies is assessed through qualitative classification prediction maps and quantitative metrics, which encompass OA, AA, Kappa, and per-class accuracy, as illustrated in Figs. 6–9 and Tables I–III. Furthermore, Tables I–III provide a comprehensive overview of the classification outcomes for CHNet and other comparative approaches on the Yancheng, Houston2013, Augsburg, and MUUFL datasets, with the superior results for OA, AA, Kappa coefficient, and per-class accuracy emphasized in bold. Our model attains the highest classification scores across all four datasets. Specifically, on the Yancheng, Houston2013, Augsburg, and MUUFL datasets, our model outperforms the second-best model in OA by 1.98%, 1.2%, 1.16%, and 0.56%, respectively. A detailed

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TABLE III
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR MUUFL DATA (BEST RESULTS ARE IN BOLD)

No.	Class (Train/Test)	CoupledCNN	S2ENET	MACN	EXViT	HCTNet	FDNet	HFSNet	M2FNet	MICF-Net	MS2CANet	ours
1	Trees (20/23186)	82.49	84.86	84.19	80.48	79.85	87.86	87.65	91.01	**97.95**	82.31	91.42
2	Mostly grass (20/4210)	70.84	74.42	78.61	64.23	70.21	77.64	75.13	66.84	64.13	**82.40**	77.83
3	Mixed ground surface (20/6822)	69.24	72.32	56.11	64.26	67.51	68.21	68.75	62.78	**91.08**	65.52	79.41
4	Dirt and sand (20/1766)	94.35	91.97	91.86	92.08	94.35	88.59	89.31	70.43	91.22	95.12	**97.18**
5	Road (20/6627)	66.59	82.11	79.81	61.42	74.35	87.51	90.21	79.45	**96.42**	73.09	87.91
6	Water (20/406)	**100**	99.77	**100**	**100**	99.55	99.78	99.55	97.08	59.89	**100**	99.10
7	Buildings shadow (20/2173)	83.64	82.60	79.98	80.93	81.74	90.74	**92.18**	76.41	67.79	86.03	80.98
8	Buildings (20/6180)	90.67	90.78	76.27	85.87	92.01	92.51	93.24	92.36	92.87	91.67	**95.98**
9	Sidewalk (20/1325)	38.02	57.94	50.62	36.34	51.86	70.84	56.55	**73.84**	51.46	44.76	52.74
10	Yellow curb (20/123)	**87.11**	82.21	78.52	77.30	85.88	80.37	68.71	67.48	16.01	86.50	83.43
11	Cloth panels (20/209)	95.98	99.59	95.98	99.19	95.98	96.39	95.58	95.98	48.58	99.59	**99.60**
	OA (%)	78.37	82.41	78.07	74.88	78.34	84.85	84.72	82.45	87.18	79.97	**87.74**
	AA (%)	79.91	83.51	79.27	76.55	81.21	85.49	83.35	79.73	70.67	82.45	**85.96**
	Kappa (%)	72.59	77.48	72.14	68.35	72.66	80.52	80.33	77.28	83.35	74.57	**84.03**

TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR AUGSBURG DATA (BEST RESULTS ARE IN BOLD)

No.	Class (Train/Test)	CoupledCNN	S2ENET	MACN	EXViT	HCTNet	FDNet	HFSNet	M2FNet	MICF-Net	MS2CANet	ours
1	Forest (20/13487)	97.42	**98.43**	97.78	94.62	95.37	98.12	96.93	81.52	96.81	96.63	97.59
2	Residential area (20/30309)	68.75	65.58	93.81	75.89	65.87	78.26	82.87	88.24	**98.21**	71.36	89.64
3	Industrial area (20/3831)	**93.08**	53.09	55.25	54.03	39.08	52.6	62.02	63.66	59.88	66.80	59.77
4	Low plants (20/26837)	75.30	80.04	87.59	81.07	87.03	85.39	83.71	86.73	**94.16**	54.43	94.13
5	Allotment (20/555)	95.86	**96.22**	93.69	81.98	94.95	82.34	89.18	75.85	25.72	93.69	94.23
6	Commercial area (20/1625)	02.15	51.45	46.64	59.51	**63.38**	58.71	41.84	52.80	24.90	43.26	60.55
7	Water (20/1510)	62.58	69.87	61.78	66.75	62.05	67.75	56.82	**75.62**	39.77	72.91	64.17
	OA (%)	75.83	75.61	88.87	79.35	76.99	82.30	83.25	84.29	87.05	69.29	**90.03**
	AA (%)	70.74	73.52	76.65	73.41	72.53	74.74	73.34	74.92	62.48	71.30	**80.01**
	Kappa (%)	68.15	67.76	84.29	71.85	69.58	76.00	76.99	77.91	82.06	59.87	**86.03**

analysis is provided for two representative datasets in the following.

1) Results on the Houston2013 Dataset: The performance of our CHNet on the Houston2013 dataset is reported in Table II. The dataset contains 15 land-cover types, with certain areas affected by cloud and fog obstruction and high scene complexity. Our model achieved an OA of 97.11% and an AA of 97.23%, both surpassing the second-place FDNet, which achieved 95.91% and 96.13%, respectively. They indicate that our method can effectively classify different land-cover types, particularly showing strong adaptability in handling complex and diverse scenes.

The experimental results further demonstrate that out of a total of 15 land-cover categories, our model achieved the best evaluation metrics in seven categories and 100% classification accuracy in five categories. Among the comparison methods, S2ENet promotes the complementary exchange of HSI-LiDAR data through a spatial enhancement module, enhancing the spatial representation of HSI features and the spectral representation of LiDAR features. This performs well in some scenes, but, due to its focus on spatial enhancement, its ability to capture spectral features is limited. ExViT, on the other hand, enhances multimodal information exchange through a cross-modal attention module, improving the discriminative ability of the class labels and, thereby, enhancing the exchange of modality information. Compared to CoupledCNN, ExViT achieved performance improvements of 13.68% and 13.35% after small-sample training.

The experimental results show that networks relying on transformer structures for feature extraction, such as MICF and ExViT, exhibit slightly weaker performance in categories with complex geometric shapes. In contrast, our proposed CHNet, which uses CNNs to extract multiscale features, demonstrates superior classification performance in categories with complex geometric shapes, especially in the "stressed grass" and "synthetic grass" categories. The primary reason for this lies in the distinct geometric shapes and sizes of objects in these categories, which require hierarchical feature extraction to effectively capture these details and variations.

Through the SRR and SR modules processing the two-modal data, our proposed network achieved 100% classification performance on categories such as "trees," "soil," "water," "tennis court," and "running track," which exhibit clear spectral and elevation information. Notably, our network CHNet achieved 100% classification accuracy in the "trees" and "water" land covers, a result that other methods failed to achieve. Fig. 7 displays the visualization of our proposed network, alongside the classification maps generated by other comparative methods for reference.

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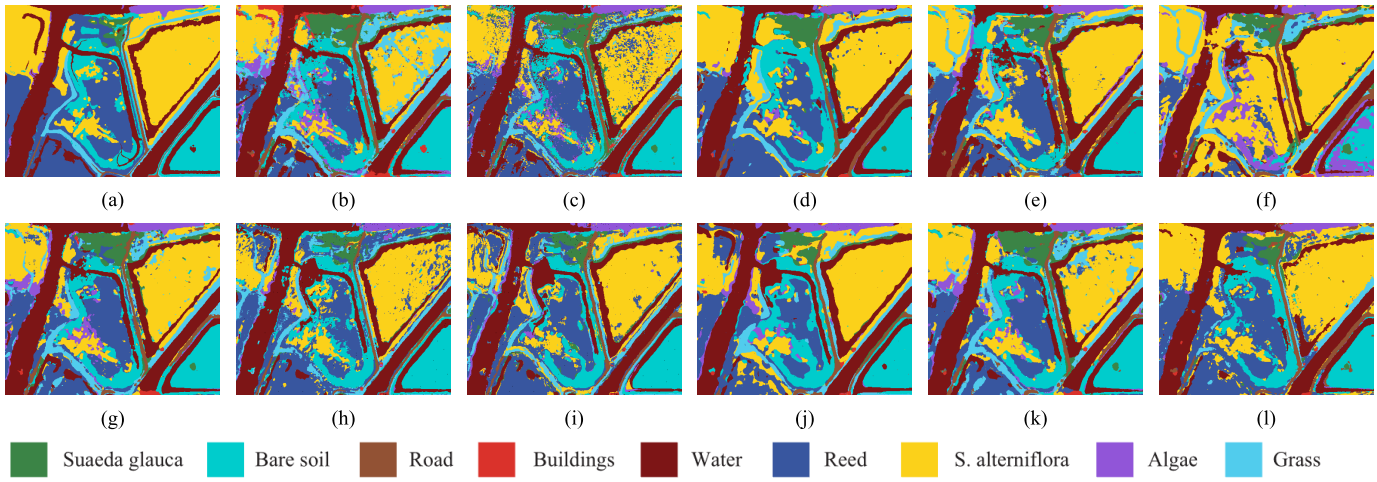


Fig. 6. Classification maps by different methods on the Yancheng dataset. (a) Ground truth. (b) CoupledCNN (67.52%). (c) S2ENet (72.90%). (d) MACN (76.02%). (e) ExViT (72.98%). (f) HCTNet (59.52%). (g) FDNet (76.65%). (h) HFSNet (73.31%). (i) M2FNet (72.97%). (j) MICFNet (75.5%). (k) MS2CANet (74.11%). (l) Ours (78.63%).

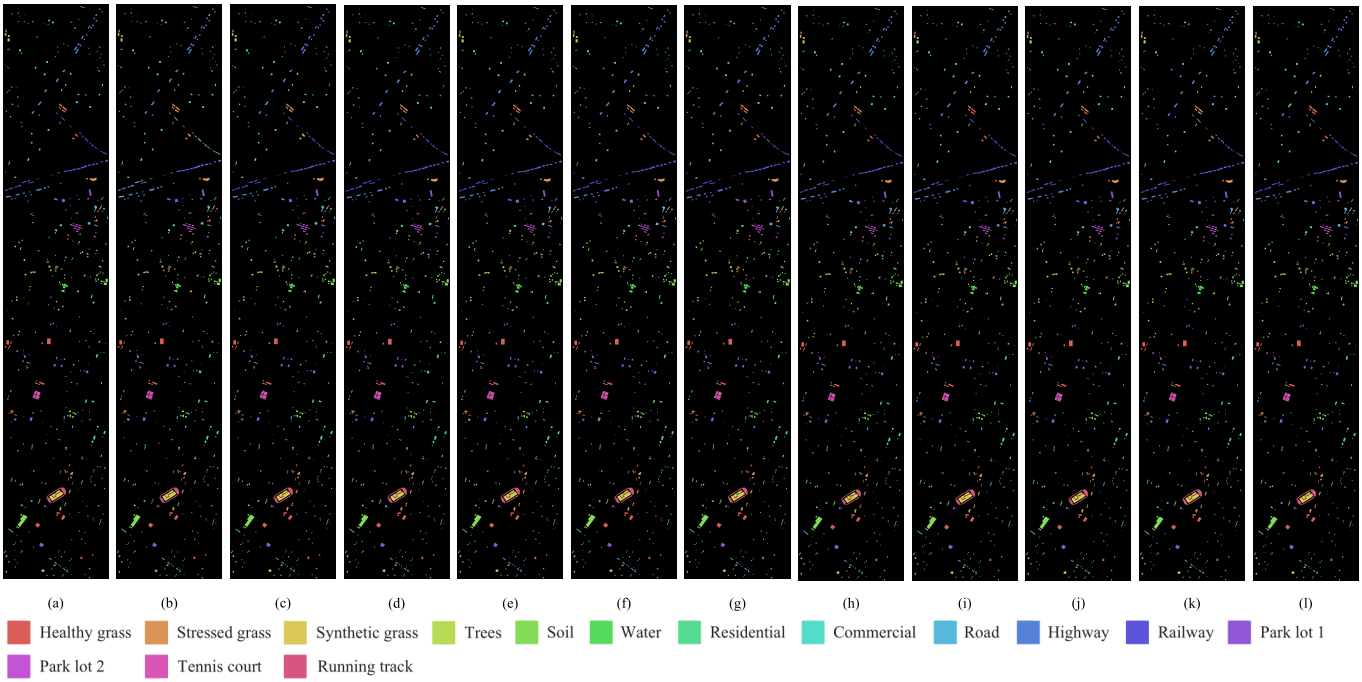


Fig. 7. Classification maps by different methods on the Houston2013 dataset. (a) Ground truth. (b) CoupledCNN (81.30%). (c) S2ENet (94.98%). (d) MACN (94.77%). (e) ExViT (94.65%). (f) HCTNet (94.53%). (g) FDNet (95.91%). (h) HFSNet (94.89%). (i) M2FNet (94.39%). (j) MICFNet (95.81%). (k) MS2CANet (88.64%). (l) Ours (97.11%).

2) Results on the Yancheng Dataset: Table I presents the classification performance of different methods on the Yancheng dataset. Unlike the Houston dataset, the Yancheng dataset exhibits a significant variation in sample sizes between categories. When training only 20 samples per category, our proposed network model performs best in two categories. In terms of overall metrics, our model achieves an OA of 78.63% and an AA of 84.13%, both of which are higher than the second-place FDNet's OA of 76.65% and AA of 83.15%.

MACN captures shallow convolutional features using an adaptive CNN encoder and combines self-attention mechanisms with convolution operations in the mixed self-attention and convolutional transformer layer, achieving local and global multiscale perception to adapt to the complexity and multiscale features of remote sensing data. FDNet employs a frequency-domain separated DWTS module to enhance the global modeling capability of high-frequency information and low-frequency information, thereby improving multimodal data fusion performance. Both methods demonstrate strong global perception capabilities, which are beneficial for understanding the structure of the entire scene or the relationships between objects, helping to improve the model's accuracy and avoid misleading local information. FDNet achieves the best classification accuracy in three categories.

For data with sample imbalance, our network utilizes the SRR and SR modules to perform redundancy reduction on

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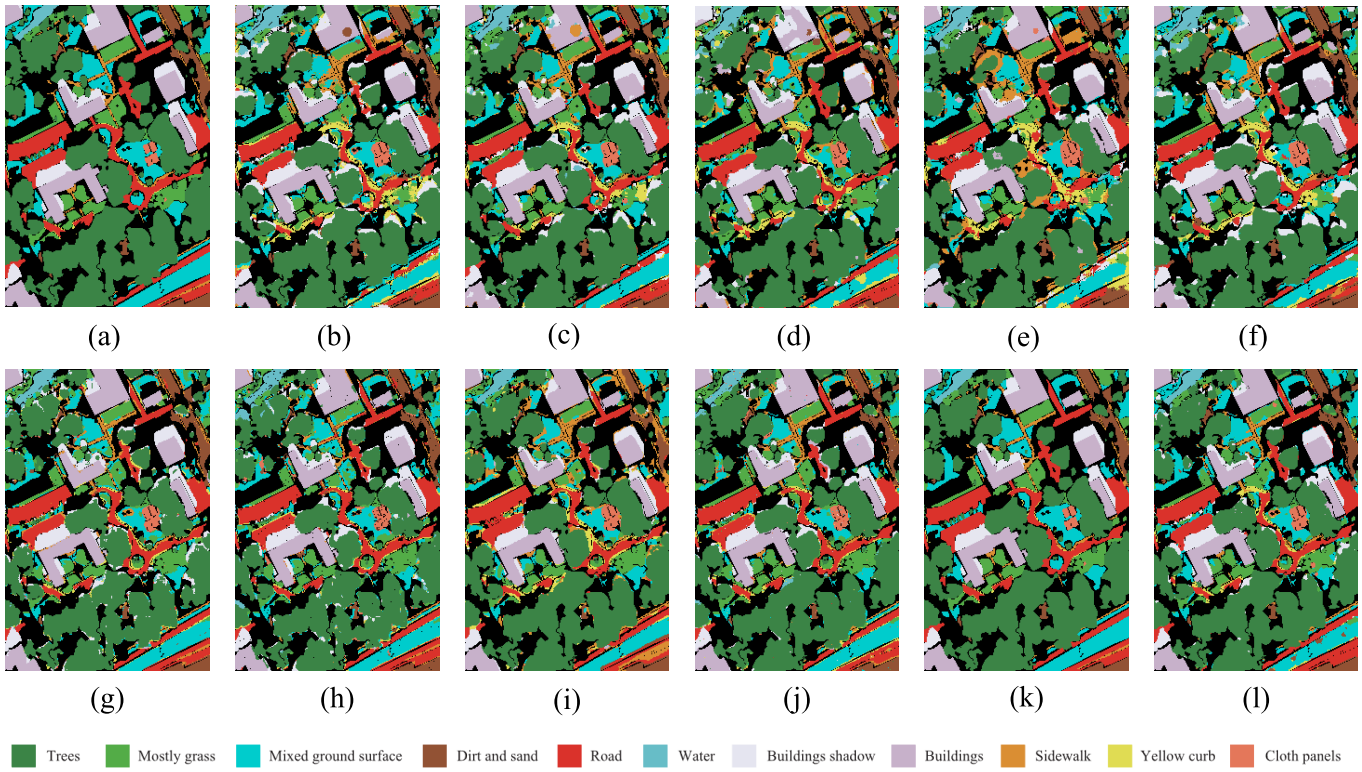


Fig. 8. Classification maps by different methods on the MUUFL dataset. (a) Ground truth. (b) CoupledCNN (78.37%). (c) S2ENet (82.41%). (d) MACN (78.07%). (e) ExViT (74.88%). (f) HCTNet (78.34%). (g) FDNet (84.85%). (h) HFSNet (84.72%). (i) M2FNet (82.45%). (j) MICFNet (87.18%). (k) MS2CANet (79.97%). (l) Ours (87.74%).

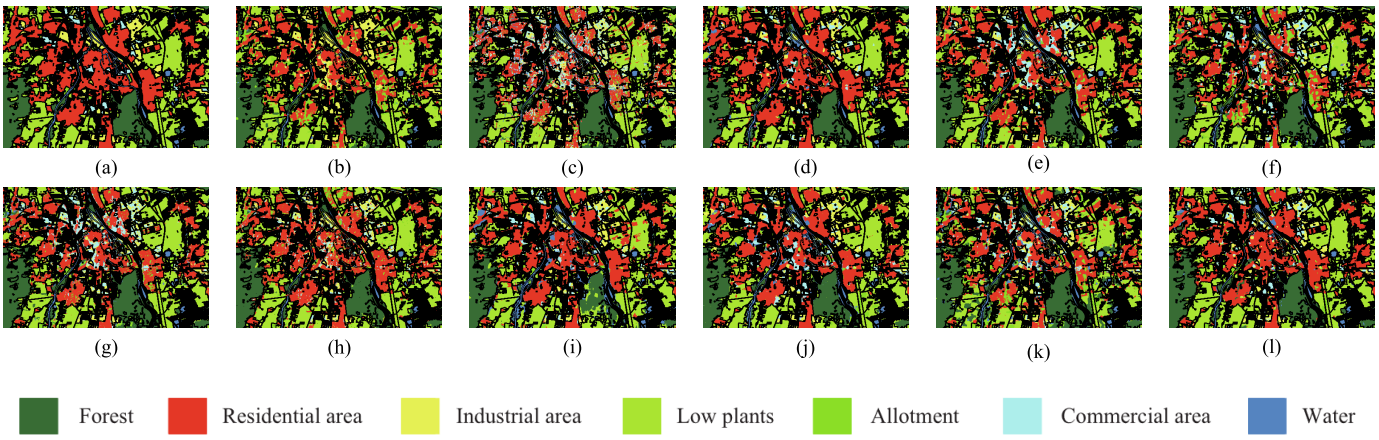


Fig. 9. Classification maps by different methods on the Augsburg dataset. (a) Ground truth. (b) CoupledCNN (75.83%). (c) S2ENet (75.61%). (d) MACN (88.87%). (e) ExViT (79.35%). (f) HCTNet (76.99%). (g) FDNet(82.30%). (h) HFSNet (83.25%). (i) M2FNet (84.29%). (j) MICFNet (87.05%). (k) MS2CANet (69.29%). (l) Ours (90.03%).

the two-modal data, removing local redundant information from a different perspective. As with the Houston dataset, for categories with clear spectral and elevation information, such as "grass" and "bare soil," our proposed network achieves the best classification accuracy. Methods such as MICF-Net perform poorly in complex regions, achieving an OA of 75.20% and an AA of only 56.87%. Due to the sample imbalance problem, this method performs well in categories with a larger sample size, such as "bare soil," "S. alterniflora," "water," and "reed" where it achieves nearly top-three classification accuracy. However, it performs poorly in regions with fewer and more complex samples. As shown in Fig. 6, MICF-Net performs well in the isolated areas of the right half, but our CHNet outperforms it in the central complex region. This further demonstrates our advantage in handling the class imbalance problem.

3) Visual Evaluation and Analysis: Figs. 6–9 present the visual classification results of several comparative methods on the Yancheng, Houston2013, Augsburg, and MUUFL datasets. Taking the Augsburg dataset in Fig. 9 as an example, it is evident that the visual classification results of the proposed model are closer to the GT, with smoother classification boundaries and overall superior classification performance. In contrast, the isolated points in the results of other

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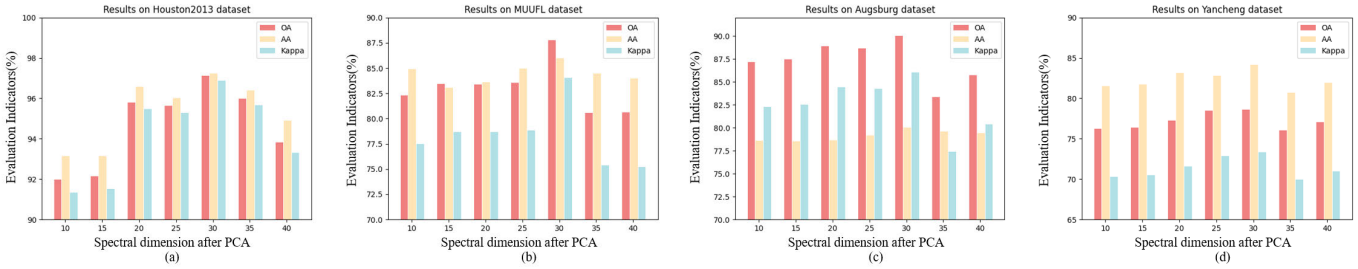


Fig. 10. Impact of different spectral dimensions after PCA on the classification performance. (a) Houston2013. (b) MUUFL. (c) Augsburg. (d) Yancheng.

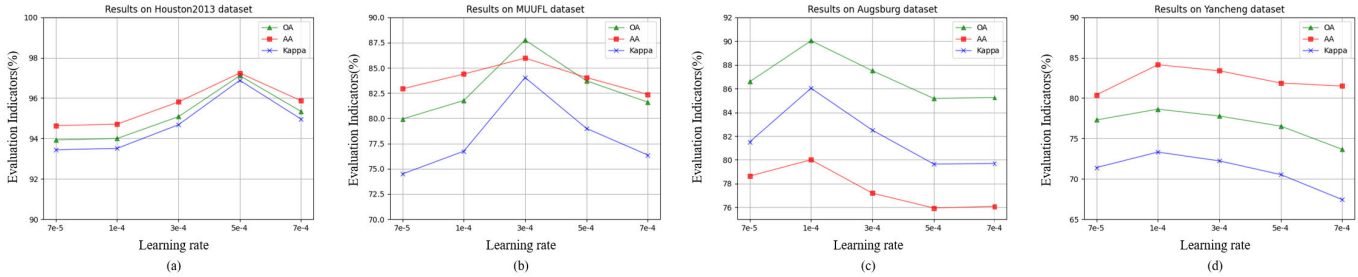


Fig. 11. Impact of different learning rates on the classification performance. (a) Houston2013. (b) MUUFL. (c) Augsburg. (d) Yancheng.

comparative methods are more pronounced. Through more effective fusion modules, our method can more accurately delineate the different categories in the scene, reducing noise in the visual classification outcomes. Notably, the classification results of our method are much closer to the real image, producing clearer classification boundaries and a significant reduction in misclassified samples.

Finally, our model demonstrates exceptional classification performance, with both quantitative and visual analyses thoroughly validating its advanced capabilities and effectiveness.

D. Ablation Studies

These modules are employed in our architecture, spanning from the coarse to fine stages. To evaluate their effectiveness, we conducted ablation experiments on the MFE, DFSF, SR, SRR, and FHFL modules, using three objective evaluation metrics: OA, AA, and Kappa. The ablation experiment results from four datasets are presented in Table V, and the baseline structure adopts double-branch convolution and is classified through a simple fully connected layer.

From the results, it is evident that our architecture achieves the highest classification accuracy across all datasets. For the four datasets, MFE and DFSF modules provide an improvement of 1%–5%, with the most significant enhancement observed on the Augsburg dataset. This improvement can be attributed to the multiscale feature extraction design and the fusion module, which greatly enhances the network's ability to recognize complex terrains. The SR and SRR modules focus on data redundancy reduction for the two modalities and initial feature extraction, preparing the model for subsequent multiscale feature extraction. The improvement varies depending on the spectral similarity between categories and terrain complexity in the datasets, but there is an overall enhancement across all datasets. The FHFL module also significantly improves the values of three evaluation metrics for all four datasets, with the most noticeable improvement on the Houston2013 dataset, where the OA value increased by 2.72%. This improvement is mainly due to the feature refinement and sparsification process, which strengthens the extracted multiscale features and, thereby, enhances classification accuracy.

These experimental results confirm the effectiveness of the four modules in improving the model's classification performance.

E. Parameters Analysis

All parameter settings are listed in Table VI. The analysis of specific parameters is given as follows.

1) Spectral Dimension After PCA: To assess the impact of varying the reduced spectral dimension b, we maintained fixed values for all other hyperparameters and tested b across a candidate set {5, 10, 15, 20, 25, 30, 35, 40}. Fig. 10 illustrates the OA, AA, and Kappa of the proposed framework across four datasets as a function of the reduced spectral dimension. The results reveal a nonlinear trend, where OA initially improves with increasing values of b, eventually stabilizing as b reaches higher values. This suggests that there is an optimal range for b, beyond which further increases do not significantly enhance performance. Taking both computational efficiency and classification accuracy into account, we determined that a reduced spectral dimension of 30 provides a favorable balance between model performance and computational demand.

2) Learning Rate: HSI and LiDAR data have significant feature differences, and HSI data emphasize spectral information, while LiDAR data provide structural and elevation information. Therefore, different learning rates may affect the model's weighting allocation between these two types of data. In the conducted experiment, the learning rate was chosen from the candidate set {7e-5, 1e-4, 3e-4, 5e-4, 7e-4}. Fig. 11

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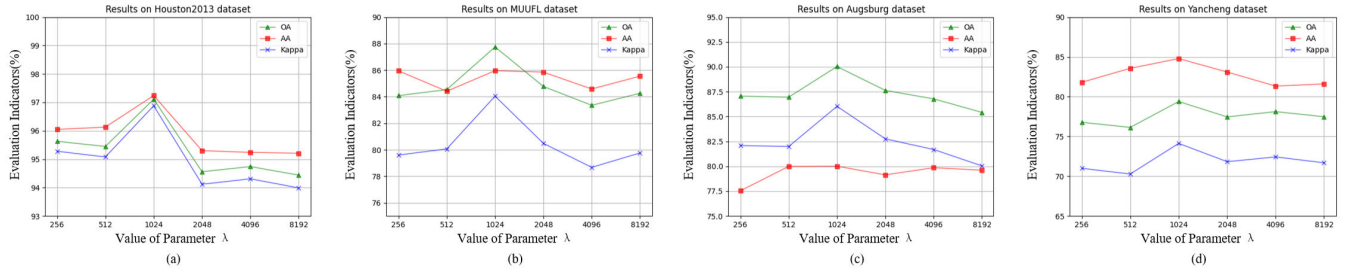


Fig. 12. Impact of different projection dimensions in the FHFL module on the classification performance. (a) Houston2013. (b) MUUFL. (c) Augsburg. (d) Yancheng.

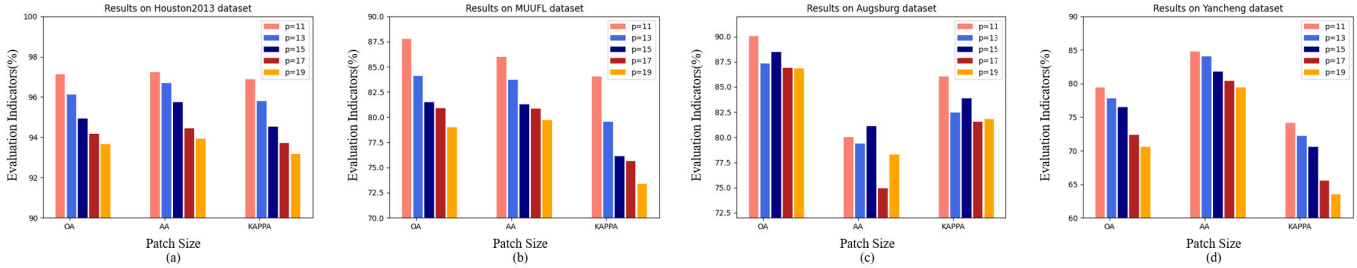


Fig. 13. Impact of different input patch sizes on the classification performance. (a) Houston2013. (b) MUUFL. (c) Augsburg. (d) Yancheng.

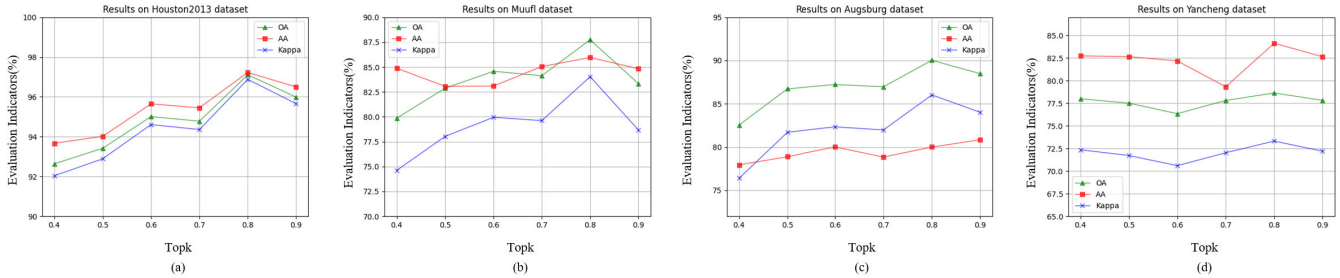


Fig. 14. Impact of different weight top-k's on the classification performance. (a) Houston2013. (b) MUUFL. (c) Augsburg. (d) Yancheng.

illustrates the OA of the proposed method across four datasets under various learning rate configurations. It is evident that the optimal learning rate for the Yancheng and Augsburg datasets is 1e-4, whereas, for the MUUFL dataset, it is 3e-4, and for the Houston2013 dataset, it is 5e-4.

3) Weight λ for Projection in the FHFL Module: The weight λ for projection in the FHFL module is a parameter used for downsampling the input features. Keeping other parameters constant, we increased the value of λ from 256 to 8192 in increments of two on four datasets. As shown in Fig. 12, when λ is too small, the model's classification performance is unsatisfactory. When λ is relatively large, the number of parameters increases, resulting in longer training times, and the classification results do not show significant improvement—in fact, they even decrease. Therefore, setting λ to 1024 yielded the best classification results for the model, and thus, we chose $\lambda = 1024$.

4) Size of Input Patch: The size of the input patch influences the extent of spatial contextual information available for each sample, which, in turn, has a direct impact on the model's capacity to extract spatial features across various scales. We conducted a comparative analysis of the effects of varying input patch sizes on the performance of classification tasks. Due to the multiscale design of the feature extraction module, the minimum input patch size was established at 11. The experimental parameters were set as {11, 13, 15, 17, 19}, with the corresponding metric values represented by five different colors. As shown in Fig. 13, the input patch size of 11 achieved the highest scores across three evaluation metrics in four datasets. Therefore, we selected an input patch size of 11 to achieve the best classification model.

5) Weight Top-k for SRR Module: The parameter top-k is the most critical hyperparameter in the SRR module, as it directly influences the effectiveness of redundancy removal and feature preservation in hyperspectral data. To investigate its impact, we conducted a comparative analysis by varying the proportion of retained data. Specifically, we set the top-k parameter to {0.4, 0.5, 0.6, 0.7, 0.8, 0.9} and evaluated its performance across four datasets. Fig. 14 illustrates the comparative metrics for different data retention ratios. The results clearly indicate that 0.8 achieves the optimal balance between redundancy removal and feature preservation, making it the best choice for this parameter.

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TABLE V
IMPACT OF DIFFERENT COMPONENTS USED IN OUR METHOD

	Yancheng			Houston2013			Augsburg			MUUFL		
Different Module	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa
Baseline	71.21	77.75	64.20	88.68	90.02	89.26	75.24	70.47	68.85	78.62	79.57	71.99
+ MFE	74.59	81.27	68.15	92.18	93.43	91.55	81.75	71.44	74.38	80.41	81.06	74.90
+ MFE + DFSF	75.83	81.67	69.73	93.33	94.35	92.79	86.37	77.07	81.09	83.05	83.52	78.16
+ MFE + DFSF + SRR	76.45	82.26	70.40	94.39	95.05	93.94	87.39	78.40	82.50	85.11	83.98	80.72
+ MFE + DFSF+ SR + SRR	77.67	83.37	72.02	94.39	95.06	93.93	87.72	74.63	82.73	85.57	83.15	81.17
+ DFSF+ SR + SRR + FHFL	76.61	81.70	70.84	93.69	94.57	93.18	85.42	76.06	79.90	78.41	78.98	72.47
+ MFE + DFSF+ SR + SRR + FHFL	**78.63**	**84.13**	**73.32**	**97.11**	**97.23**	**96.88**	**90.03**	**80.01**	**86.03**	**87.74**	**85.96**	**84.03**

TABLE VI
PARAMETERS IN DIFFERENT MODULES

Parameters	Yancheng	Houston2013	Augsburg	MUUFL
Learning Rate	1e-4	5e-4	1e-4	3e-4
Input Patch Size	11	11	11	11
Topk Rate	0.8	0.8	0.8	0.8
Weight λ	1024	1024	1024	1024
Spectral Dimension after PCA	30	30	30	30

V. CONCLUSION

This article proposes a two-stage HSI and LiDAR joint classification architecture called CHNet, a coarse-to-fine high-order network. The network employs an SRR module and an SR module to remove redundancy in HSI and LiDAR data, thereby enhancing the quality of multisource data and its feature representation. Through a feature pyramid structure, multiscale features are extracted, and a DFSF module is developed to dynamically enhance and fuse these features across different scales. For further enhancing feature representations and dimensionality reduction, an FHFL module that considers high-order feature statistics of land covers is integrated into the network to improve final classification accuracy. Experimental results on four HSI and LiDAR datasets, in comparison with existing classification methods, confirm the superior performance and effectiveness of the proposed approach. In conclusion, our proposed CHNet enables the extraction of more discriminative information and effectively integrates heterogeneous data features for joint HSI and LiDAR classification.

The network framework presented in this article is based on the CNNs series, but it does not model the global and explicit structural features of HSI and LiDAR data. Therefore, future research will focus on network architectures that can learn global spatial–spectral features and model structures, such as the transformer and the graph convolution neural network (GCN). Moreover, optimizing the model architecture to reduce parameter counts and computational costs, thereby improving its applicability in computationally constrained scenarios, is another direction for future work.

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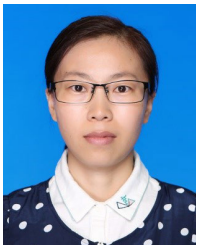
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